



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Designing a metadata codebook

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Total time: 10:45-12:00 = 1h15
Cours: 30 min
TP: 45 min

Credits: Adapted from the 'Metacoding' lecture given by Dakis-Yaoba Ouédraogo at the CESAB, October 2025.

SOUTIENS INSTITUTIONNELS





Overview

A good codebook is not a list of interesting variables — it is a set of reproducible decisions.

1 Start with the question

What do you need to be able to describe, compare or map?

2 Define the variables

Name them, define them, and specify exactly what can be entered.

3 Define the study

Decide what one row in your database represents.

4 Test & refine

Pilot ambiguous variables before full extraction.

Why do it?

A systematic map can describe the quantity and nature of evidence — and reveal clusters and gaps.

Question

What has been studied?

population • intervention • outcome

How much?

counts by category

The codebook turns papers into a comparable evidence map

	Restoration	Protection	Management	Mixed
Local	18	7	11	4
National	9	14	6	8
Regional	3	5	12	10

The payoff

You can see where evidence is concentrated — and where a future systematic review/meta-analysis may be feasible.

Overview: deciding what to code

Research question

Primary + secondary questions

What do you need to know about the distribution of evidence to answer them?

Study characteristics

PICO/PECO elements can be a good place to start to characterise the distribution of evidence.

E.g.:

Population, context, design, intervention/exposure, outcomes

Purpose test

“This variable will allow us to [analysis you want to do/figure you want to make].”

If you cannot finish it, reconsider.

What should you code?

Examples of types of information to code. But exact variables **depend on research question**.

Bibliographic

year • journal • language • document type
-- often available as exported metadata
from citation indexed databases

Study design

experimental • observational • modelling •
survey

Population / system

taxon • ecosystem • demographic group

Context

country • region • setting • spatial scale

Exposure / intervention

what was done / measured / compared

Outcomes

what response was assessed • level of
organisation

What NOT to code?

The trap

“That sounds interesting — let’s add it.”

The consequence

More variables → more time → more data you may never use.

OR

You don’t code the variables that are **ACTUALLY** important.

Use the PURPOSE TEST

Know your research question/key messages/hypotheses beforehand.

For every variable, be able to finish:

“This variable will allow us to [analysis/figure to support key message].”

If you cannot complete the sentence, reconsider the variable.



How to define a variable

VARIABLE

What are you coding?

INSTRUCTION

What exactly should the coder look for?

TYPE

What can be entered?

VALUES

What labels are allowed? And **how is each value defined?**

EXAMPLES

What does a real entry look like?

How to define a variable

Example: 'nature-based intervention'

BAD CODEBOOK

intervention_type

restoration | protection | management | mixed

What counts as restoration? What about reforestation, ecological restoration, or a lab experiment mimicking restoration?

GOOD CODEBOOK

intervention_type *What broad type of nature-based intervention are considered in the study?*

restoration: the intervention aims to return degraded/damaged ecosystems toward a pre-disturbance state. Define inclusions, exclusions and edge cases.

protection: ...

...

The variable is defined, along with the meaning of each label



Choose the data type

Structure the field so the database can enforce your decision.

Select one

one category

Select multiple

several categories

Boolean

yes / no

Integer

whole number

Text

free text when necessary

Date

YYYY-MM-DD

Be as precise and standardised as possible.

“France” ≠ “French” ≠ “FR”

Validation is part of codebook design

Prevent inconsistency at the point of entry.

Instead of cleaning this...

USA

United States

US

...build the allowed values into the field.

This can be done in google sheets easily using 'Data validation' or in many other online coding platforms

country

Select one ▼

France

Germany

United Kingdom

United States

What is a 'study'?

Define the conditions that make one unique row in your database.

ONE PUBLICATION

Intervention A

→ outcome 1

Intervention B

→ outcome 2

Could be 2
studies

OR

Site 1

restoration → outcome 1

Site 2

restoration → outcome 2

If these differ in a way that matters to your analysis (e.g. an effect modifier), they may need separate rows.

Agree this BEFORE coding

Example: one paper can have many studies

Hédouin et al. (2016)

$$1 \text{ species} \times 2 \text{ exposures} \times 9 \text{ outcomes} = 18 \text{ studies}$$

Paper-level fields

author • year • DOI • document type

Study-level fields

taxon • exposure • outcome • location

Why it matters

One paper ≠ one row by default.

Decide how to handle missing information

Not stated

The article does not report the information.

Not applicable

The variable genuinely does not apply.

Author contact

Information may be recoverable by contacting authors.

Linked study

Use a related publication only when the linkage is defensible.

Unclear

Information is present but difficult for the reviewer to interpret with sufficient precision to classify

Pilot testing: where the codebook reveals its weaknesses

If a variable looks obvious on paper, try coding real studies.

Example

variable: spatial scale

At what spatial scale was the study conducted?

local	sub-national	national	regional	global
municipality	larger than local	country	multiple countries	full global

Now imagine: multiple local sites within a country, and the level of inference was country-level, but the scale of data collection was local.

Pilot on a small sample

1. Read the codebook
2. Code independently
3. Flag ambiguous cases
4. Compare decisions
5. Revise definitions / values
6. Repeat if needed



Pilot testing/reviewer training: consistency

Iteratively design & refine the codebook with your review team so that independent reviewers make consistent decisions

Two coders

Both independently code the same sample.

Compare

Identify disagreements and discuss why they happened.

Refine codebook

Clarify definitions, add examples, adjust categories.

Example of a codebook

Inspiration: look at published protocols of similar syntheses in your field

VARIABLE DESCRIPTIONS						
Information category	Variable name	Instructions	Variable type	Values allowed	Example 1	Example 2
Intervention	intervention type	What broad type of nature-based intervention are considered in the study	select one	restoration, protection, management, created ecosystems, mixed created/non-created ecosystems	restoration	mixed

[Link to codebook template](#)

VALUE DESCRIPTIONS			
Information category	Variable name	Codes	Description
Intervention	intervention type	restoration	Code as restoration if the intervention is described as active or passive restoration aiming to return degraded, damaged, or destroyed ecosystems to a pre-disturbance natural state. The intervention may be called a synonymous term such as ecological restoration; functional restoration; ecosystem restoration; structural restoration; intervention ecology; reclamation; reforestation; rehabilitation; revegetation. Note however that the use of these terms did not necessarily mean they met our definition of restoration. We relied on the intervention description, not what it was called. Given the description, these terms may in fact have been coded as created ecosystems (see below) or were not a relevant nature-based intervention as per our scope at all. Also include cases where the intervention is a lab-based experiment mimicking restoration.
		protection	Code as protection if the described intervention involves marine, land, or coastal site-specific protection, e.g. protected areas and their management, private land conservation measures, reserves, conservancies, areas protected by indigenous or local communities (e.g. sacred sites), and locally managed marine areas (LMMA) with specific set-aside 'conservation zones'.
		management	Code as management for natural resource management which do not include restoration or protection measures. E.g. ecosystem-based fire management, species management (e.g. for ecotourism), actions characterized as forestry or forest management where there is an explicit reference to the importance of species diversity, often framed as 'close-to-nature' (e.g. for timber production or production of other ecosystem goods and services). Any intervention reported as forest management (e.g. salvage logging) is included provided the study produces outcomes relevant for understanding how the management intervention addresses an existing or anticipated climatic impact.
		combination	Code as combination when more than one type of intervention involving natural / semi natural ecosystems (management, restoration, or protection) is used. E.g. natural revegetation/reforestation in set asides/exclosures to protect abandoned land for regeneration, or when more than one intervention takes place over a landscape, and the climate impact outcomes result from the combined effects of each intervention.

Summary codebook development checklist

- ✓ Every variable has a clear purpose linked to the research question.
- ✓ Each variable has an operational definition and clear allowed values.
- ✓ Labels are defined — including edge cases and exclusions.
- ✓ The unit of analysis / definition of a study is explicit.
- ✓ Missing-data rules are documented.
- ✓ Data types and validation reduce inconsistent entry.
- ✓ The codebook has been pilot-tested on real studies.
- ✓ Coder disagreements have been discussed and used to refine the rules.



Activity

Total time: 45 min, rough time guidelines given

- In same groups you used to brainstorm your figure
- Select 1 variable needed to make your figure (e.g., study location, spatial scale, study type, etc.)
- [~15 min] Group: Design the codebook for that variable (variable definition, type, label definitions). You may use the template provided, or your own choice of template/online platform
- [~15 min] Individual: Every person picks a paper from their own field they know well and attempts to code that variable for their paper.
- [~15 min] Group: Then discuss between your group. Describe the study and how you coded your variable – do the others agree with your choice? Did you run into any edge cases that were difficult to categorise? Discuss possible refinements.